SANDRINGHAM COLLEGE

Student Name _____

Teacher Name _____

Checkpoint Task

8 SEAL Mathematics

ALT 5 – Sketching Graphs

- Reading time is 5 minutes
- Writing time is 80 minutes

Approved materials for this part

- Protractors, set squares, and aids for curve sketching
- One scientific calculator
- One bound reference

Materials supplied

- Question and Answer Book

Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the room.

Contents

13 questions (50 marks)

Instructions

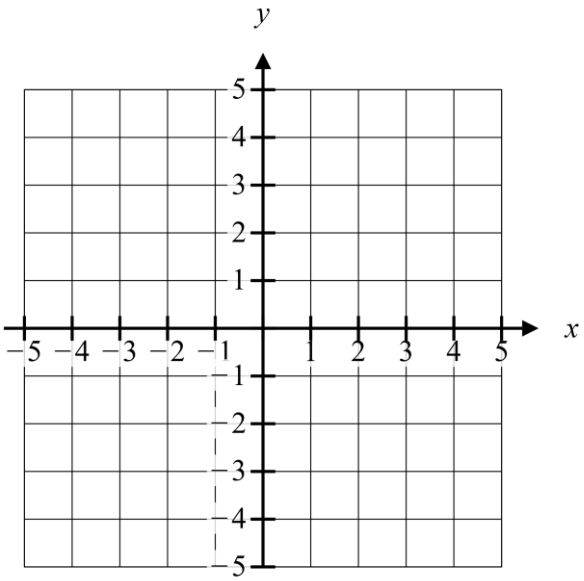
- Answer **all** questions in the spaces provided.
- Write all your responses in English.
- Unless otherwise specified, an **exact** answer is required for each question.
- In questions where more than one mark is available, appropriate working **must** be shown.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.

Question 1 (1 + 1 + 2 + 2 = 6 marks)

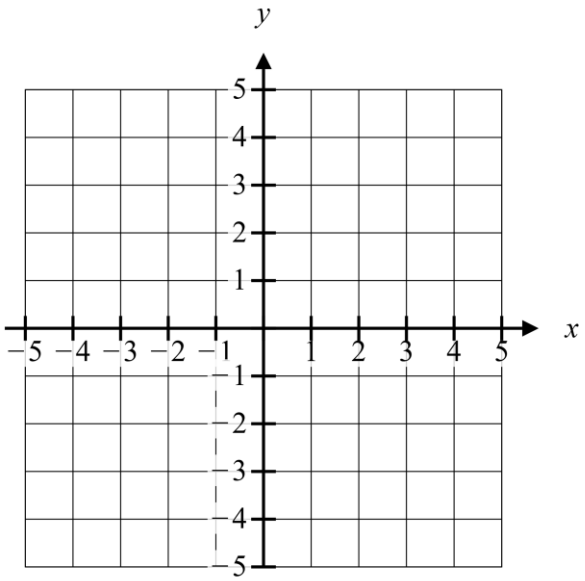
Sketch the following lines.

Label axis intercepts with their coordinates.

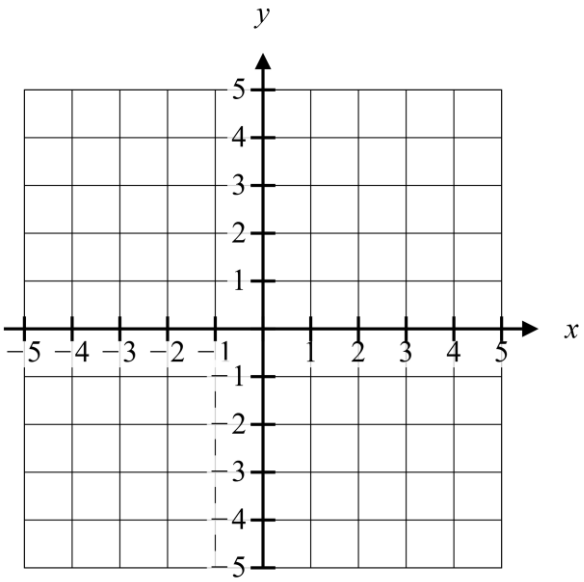
a. $x = \frac{5}{2}$



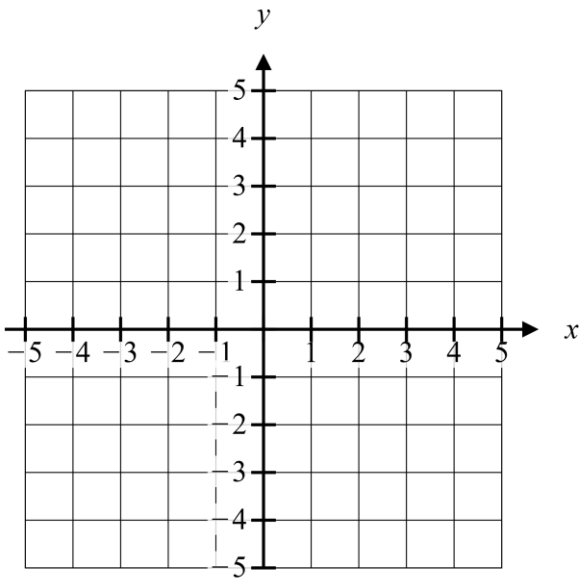
b. $y = \frac{5}{2}x$



c. $y = -\frac{2}{3}x + 1$



d. $5y + 6x = 12$



Question 2 (5 marks)

A line passes through the points $A(1, -3)$ and $B(-4, 7)$.

a. Find the midpoint of A and B. 1 mark

b. Find the distance between A and B. 2 marks

c. Determine the equation of the line through A and B.
Write your answer in the form $y = mx + c$. 2 marks

Question 3 (1 mark)

A line has gradient 5 and passes through the point $(4, -1)$.

State the equation of the line. 1 mark

Question 4 (2 marks)

A line, L, is given by the equation $y = \frac{11}{5}x - 6$.

Determine the equation of the line perpendicular to L which passes through $(3, 2)$. 2 marks

Question 5 (2 marks)

Determine if the following two lines are parallel, perpendicular, or neither. 2 marks

$$-5x - 2y = 1, \qquad 4x - 10y = 7$$

Question 6 (3 marks)

Find the coordinates of the point of intersection between the following two lines. 3 marks

$$3x - 15y = 1$$
$$2x + 10y = 4$$

Question 7 (15 marks)

For each of the following quadratics (in parts **a.**, **b.**, and **c.**) find the **coordinates** of the:

- **i.** x-intercepts,
- **ii.** y-intercept,
- **iii.** turning point.

a. $y = 10(x + 5)(x - 3)$. 5 marks

b. $y = 4x^2 - 8x + 1$ 5 marks

c. $y = 2(x + 4)^2 - 50$ 5 marks

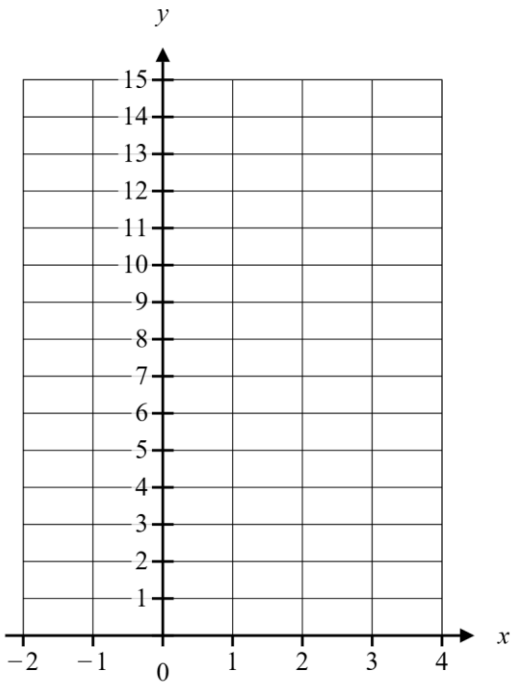
Question 8 (5 marks)

Consider the quadratic $y = 2x^2 - 4x + 5$

a. Complete the table of values below. 1 mark

x	-1	0	1	2	3
y					

b. Hence, sketch the graph of $y = 2x^2 - 4x + 5$ for $-1 \leq x < 3$.
Label end-points with their coordinates. 2 marks



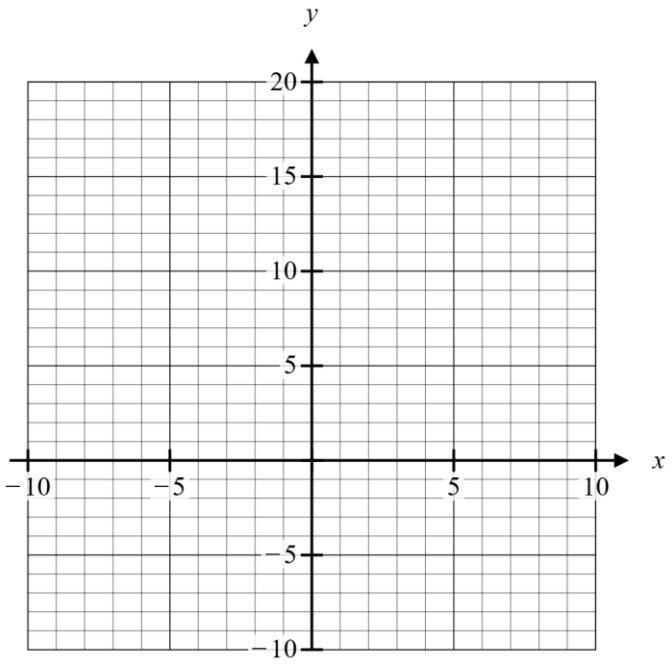
c. Find the **exact** values of x such that the line $y = 8$ crosses the quadratic above. 2 marks

Question 9 (1 mark)

A parabola with equation $y = -(x - 3)(x + 5)$ has the features in the table below.

Feature	Coordinates
x-intercepts	$(-5, 0)$ and $(3, 0)$
y-intercept	$(0, 15)$
turning point	$(-1, 16)$

Sketch the graph of the parabola. 1 mark



Question 10 (2 marks)

A parabola has a turning point at $(-3, 10)$ and passes through $(2, -5)$.

Determine the equation of the parabola in turning point form. 2 marks

Question 11 (2 marks)

A parabola passes through $(-5, 0)$ and $(7, 0)$ and passes through $(-2, 18)$.

Determine the equation of the parabola in factorised form. 2 marks

Question 12 (3 marks)

For each point, apply the corresponding transformation.

a. Translate $(5, 1)$ three units right. 1 mark

b. Dilate $(12, -6)$ by factor $\frac{2}{3}$ from the y-axis. 1 mark

c. Reflect $(11, 22)$ in the x-axis. 1 mark

Question 13 (3 marks)

For each equation, apply the corresponding transformation.

a. Translate $y = x^2$ three units up. 1 mark

b. Dilate $y = x^2$ by factor 4 from the x-axis. 1 mark

c. Dilate $y = x^2$ by factor $\frac{3}{5}$ from the y-axis. 1 mark